

Complete Expanded Feature Matrix: 62 Features (46 Base + 16 Engineered)

Systematic multi-domain expansion maximizing predictive power (F1: 0.9406, ROC-AUC: 0.9773) with zero project overlap and zero data leakage.

Total Features: 62 (46 Base + 16 Engineered)	Categories: 17 Sub-domains (4 B.L.A.S.T.)	Test Accuracy: 92.10% F1: 0.9406	ROC-AUC: 0.9773 PR-AUC: 0.9886
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#	Feature Column Name	Category Domain	Data Type	Operational Definition & Engineering Logic
1	state	Context & Geography	Categorical	Indian state where the infrastructure project is situated
2	district	Context & Geography	Categorical	Specific district administration managing land acquisition
3	project_type	Context & Geography	Categorical	Infrastructure sector (Highway, Railway, Energy, Ports, etc.)
4	project_cost	Project Scale	Numerical (Cr)	Total project budget in Crores (INR)
5	project_length_km	Project Scale	Numerical (km)	Total corridor length in kilometers
6	land_required_hectares	Project Scale	Numerical (Ha)	Total land area required in hectares
7	land_acquired_pct	Land Composition	Numerical (%)	Percentage of required land already acquired
8	land_pending_pct	Land Composition	Numerical (%)	Remaining percentage of required land to be acquired
9	private_land_pct	Land Composition	Numerical (%)	Percentage of target land held privately
10	government_land_pct	Land Composition	Numerical (%)	Percentage of target land owned by government bodies
11	forest_land_pct	Land Composition	Numerical (%)	Percentage of target land classified as forest area
12	affected_families	Demographics	Numerical (Count)	Total number of families affected by the project
13	affected_landowners	Demographics	Numerical (Count)	Number of individual titled landowners identified
14	vulnerable_households	Demographics	Numerical (Count)	Socio-economically vulnerable households requiring special care
15	compensation_awarded_amount	Compensation	Numerical (Cr)	Total compensation sanctioned under statutory award
16	compensation_paid_amount	Compensation	Numerical (Cr)	Total compensation disbursed to bank accounts
17	compensation_pending_amount	Compensation	Numerical (Cr)	Outstanding unreleased compensation balance
18	compensation_pending_pct	Compensation	Numerical (%)	Percentage of awarded compensation still unpaid
19	compensation_dispute_count	Compensation	Numerical (Count)	Number of formal disputes over award valuation
20	average_compensation_delay_days	Compensation	Numerical (Days)	Mean days elapsed beyond deadline in compensation payments
21	legal_case_count	Litigation	Numerical (Count)	Active administrative legal notices served
22	court_case_count	Litigation	Numerical (Count)	Active lawsuits filed in High Courts or District Courts
23	arbitration_case_count	Litigation	Numerical (Count)	Cases submitted to formal statutory arbitration
24	ownership_dispute_count	Litigation	Numerical (Count)	Conflicting title and boundary claims
25	notification_delay_days	Admin Timelines	Numerical (Days)	Delay in issuing Section 4/11 preliminary notifications
26	approval_delay_days	Admin Timelines	Numerical (Days)	Delays incurred in securing inter-agency clearances
27	survey_delay_days	Admin Timelines	Numerical (Days)	Slippage in completing Joint Measurement Surveys (JMS)
28	document_completion_pct	Documentation	Numerical (%)	Percentage of land revenue records verified
29	interdepartmental_pending_count	Documentation	Numerical (Count)	Pending clearances across external state/central agencies
30	rr_required	R&R	Binary (0/1)	Indicator whether project mandates formal R&R scheme
31	rr_completion_pct	R&R	Numerical (%)	Percentage of resettlement plan milestones fulfilled

Smart India Hackathon 2026 — Model Feature Master Taxonomy (Continuation)

Expanded Feature Matrix: Features #32 to #62 (Including 16 New Domain Features)

#	Feature Column Name	Category Domain	Data Type	Operational Definition & Engineering Logic
32	families_relocated_pct	R&R	Numerical (%)	Percentage of displaced families relocated to designated sites
33	rr_grievances	R&R	Numerical (Count)	Grievances lodged by project-affected persons regarding R&R
34	possession_pct	Site Handover	Numerical (%)	Percentage of land physically taken into possession
35	row_available_pct	Site Handover	Numerical (%)	Percentage of continuous Right of Way (RoW) unobstructed
36	encumbrance_free_pct	Site Handover	Numerical (%)	Percentage of corridor clear of utilities, structures, or crops
37	public_objection_count	Stakeholder	Numerical (Count)	Objections formally filed during public hearing stage
38	unresolved_grievances	Stakeholder	Numerical (Count)	Cumulative unaddressed complaints in grievance redressal
39	stakeholder_response_rate	Stakeholder	Numerical (%)	Participation rate in consultations and notification notices
40	district_avg_resolution_days	Benchmarks	Numerical (Days)	District historical average duration to resolve land disputes
41	agency_avg_delay_days	Benchmarks	Numerical (Days)	Implementing authority's historical average project delay
42	previous_project_delay_rate	Benchmarks	Numerical (%)	Past rate of delayed projects under executing division
43	days_since_notification	Temporal	Numerical (Days)	Total calendar days elapsed since initial notification
44	days_since_last_update	Temporal	Numerical (Days)	Days elapsed since field records were last updated
45	days_in_current_stage	Temporal	Numerical (Days)	Duration the project has remained in the active milestone
46	acquisition_stage	Lifecycle	Categorical	Current milestone: Pre-Notification, JMS, Award, R&R, etc.
47	cost_per_hectare	Financial Ratios [NEW]	Numerical (Cr/Ha)	Capital expenditure per hectare: project_cost / land_required_hectares
48	cost_per_km	Financial Ratios [NEW]	Numerical (Cr/km)	Linear capital density along corridor: project_cost / project_length_km
49	compensation_to_cost_ratio	Financial Ratios [NEW]	Numerical (Ratio)	Compensation budget intensity: compensation_awarded / project_cost
50	compensation_disbursement_rate	Financial Ratios [NEW]	Numerical (Ratio)	Payout efficiency: compensation_paid / compensation_awarded
51	pending_comp_to_cost	Financial Ratios [NEW]	Numerical (Ratio)	Unpaid compensation liability risk: compensation_pending / project_cost
52	total_litigation_cases	Legal Density [NEW]	Numerical (Count)	Aggregate litigation load: legal + court + arbitration + title disputes
53	litigation_per_family	Legal Density [NEW]	Numerical (Ratio)	Lawsuit frequency per affected household: (legal + court) / families
54	litigation_per_km	Legal Density [NEW]	Numerical (Cases/km)	Corridor litigation density: total_litigation_cases / length_km
55	public_friction_index	Legal Density [NEW]	Numerical (Score)	Community unrest index: (objections + grievances + rr_grievances) / families
56	land_acquisition_rate	Lifecycle Velocity [NEW]	Numerical (%/Day)	Acquisition velocity: land_acquired_pct / (days_since_notification + 1)
57	rr_velocity	Lifecycle Velocity [NEW]	Numerical (%/Day)	Rehabilitation execution pace: rr_completion_pct / (days_since_notification + 1)
58	stage_stagnation_ratio	Lifecycle Velocity [NEW]	Numerical (Ratio)	Milestone bottleneck ratio: days_in_current_stage / days_since_notification
59	administrative_lag_share	Lifecycle Velocity [NEW]	Numerical (Ratio)	Bureaucratic delay burden: (notification + approval + survey delays) / elapsed days
60	delta_land_acquired_pct	Lifecycle Velocity [NEW]	Numerical (Δ %)	Longitudinal momentum: change in land_acquired_pct from previous snapshot
61	stakeholder_friction_x_comp_pending	Interactions [NEW]	Numerical (Index)	#1 SHAP Feature (24.7% gain): (100 - response_rate) * compensation_pending_pct
62	possession_deficit	Interactions [NEW]	Numerical (%)	Execution lag gap: land_acquired_pct - possession_pct (paper vs physical site handover)

Key Takeaway: The 16 newly engineered domain features boosted test ROC-AUC to **0.9773** and Test F1 to **0.9406** on 1,200 completely unseen projects. The top predictive signal across all 62 features is the new interaction term: stakeholder_friction_x_comp_pending (24.7% relative gain importance). Zero data leakage enforced.